

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper No. BPT-401: Electronics and Modern Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **five** of the following:

[5 × 4 = 20]

- What are polar and non-polar dielectrics? Explain the terms electronic, ionic, and dipolar polarizabilities.
- Electrons are accelerated in a TV tube through a potential difference of 10 kV. Find the highest frequency of the electromagnetic wave emitted when these electrons strike the screen of the tube. Justify that these waves are X-rays.
- Explain the various parameters in the given equation for a p-n junction diode:

$$I = I_0 \left[e^{\frac{eV}{\eta kT}} - 1 \right]$$

Prove that the dynamic resistance of the diode is given by $r_d = \frac{\eta kT}{eI}$.

- How is current gain α different from β ? Explain with expressions. Determine the current gain β and emitter current I_E for a transistor circuit in CE configuration, where $I_B = 50 \mu\text{A}$ and $I_C = 3.65 \text{ mA}$.
- Show that the de Broglie wavelength of a particle of mass m , charge q , and accelerated through a potential difference of V volts is:

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

- Explain the important features of paramagnetic, diamagnetic, and ferromagnetic materials. Give the domain theory of ferromagnetism.
- Explain why a transistor needs to be biased. Draw the load line and explain its importance.

Question 2.

- Explain the formation of the depletion region in a P-N junction diode. Find the expression for the width of the depletion region and the height of the potential barrier without external voltage. [10]
- A silicon p-n junction has a depletion width of $20 \mu\text{m}$ at a reverse voltage of 16 V. If the reverse voltage is increased to 25 V, find the new depletion width. [4]

Question 3.

- (a) What is Lorentz local field? Derive the Lorentz formula for the internal electric field. Hence, derive the Clausius-Mossotti equation:

$$\frac{K - 1}{K + 2} = \frac{N\alpha_e}{3\epsilon_0}$$

where the symbols have their usual meanings. [10]

- (b) A slab of dielectric material of dielectric constant 10 and thickness 5 mm is placed between the parallel plates of a capacitor. If a potential difference of 10^5 V is applied between the plates, evaluate the Lorentz field. [4]

OR

- (a) Define magnetic induction $\vec{B}$, magnetization $\vec{M}$, magnetic susceptibility χ_m , and permeability μ . Establish the relationship between them. [5]
- (b) Obtain Langevin's formula for the dipolar polarizability of a polar dielectric substance. [5]
- (c) Evaluate the Larmor frequency of an electronic orbit when it is subjected to a magnetic induction field of 2 Tesla. (Given $e = 1.6 \times 10^{-19}$ C and $m = 9.1 \times 10^{-31}$ kg). [4]

Question 4.

- (a) Distinguish between continuous and characteristic X-rays. Show the plot of X-ray intensity versus frequency observed in both cases. [4]
- (b) Derive Bragg's law for X-ray diffraction from a layer of atoms in a plane. [6]
- (c) X-rays of wavelength $0.71 \text{ \AA}$ are reflected from the (110) plane of a rock salt crystal ($d = 2.82 \text{ \AA}$). Calculate the glancing angle θ corresponding to the second-order Bragg spectrum. [4]

OR

- (a) Define mobility of charge carriers in a semiconductor. Find the expressions for the drift current and diffusion current in a semiconductor. [5]
- (b) What are the various transistor biasing techniques? Draw the circuit diagram of a self-bias circuit and describe its working for stabilization. [5]
- (c) What is rectifier efficiency? Show that the efficiency of a half-wave rectifier is:

$$\eta = \frac{40.6\%}{1 + R_f/R_L}$$

where R_f is the forward resistance of the diode and R_L is the load resistance. [4]

Question 5.

- (a) Draw the circuit diagram of a single-stage RC coupled common-emitter amplifier. Explain the function of each resistance and capacitance used in the circuit. Plot the frequency response of the amplifier. [6]
- (b) What is the photoelectric effect? Explain photoelectric current and retarding potential. The photoelectric threshold wavelength for a metal is $3000 \text{ \AA}$. Find the kinetic energy of an electron ejected from it by radiation of wavelength $1200 \text{ \AA}$. (Given $h = 6.62 \times 10^{-34}$)
- (b) Explain active region, cut-off region, and saturation region in transistor operations using suitable circuit characteristics and curves. [2]